

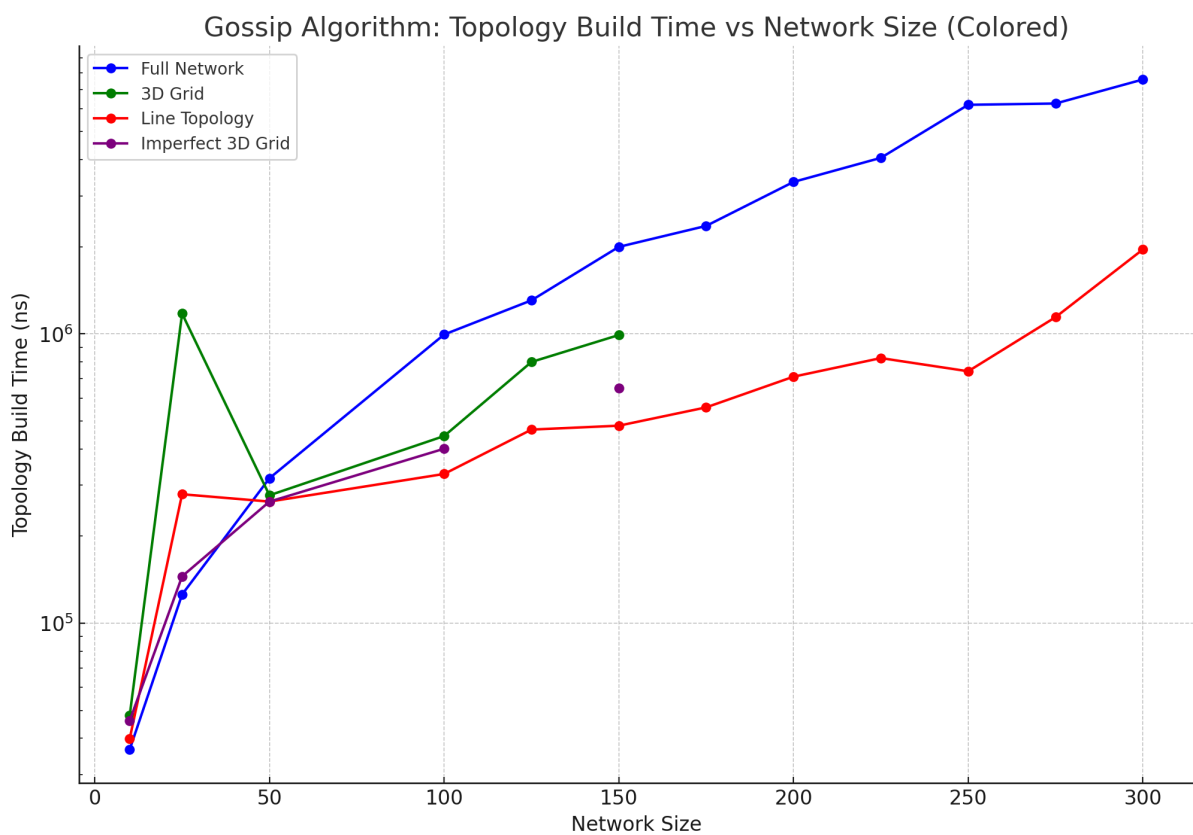
Convergence Analysis of Gossip and Push-Sum Algorithms

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This is Gossip Algorithm Graph:



Findings from the Gossip Algorithm Graph:

Full Network Topology:

Convergence times are highly variable, increasing unpredictably with network size due to full connectivity overhead.

3D Grid Topology:

Shows moderate, consistent convergence times; structure aids predictable propagation.

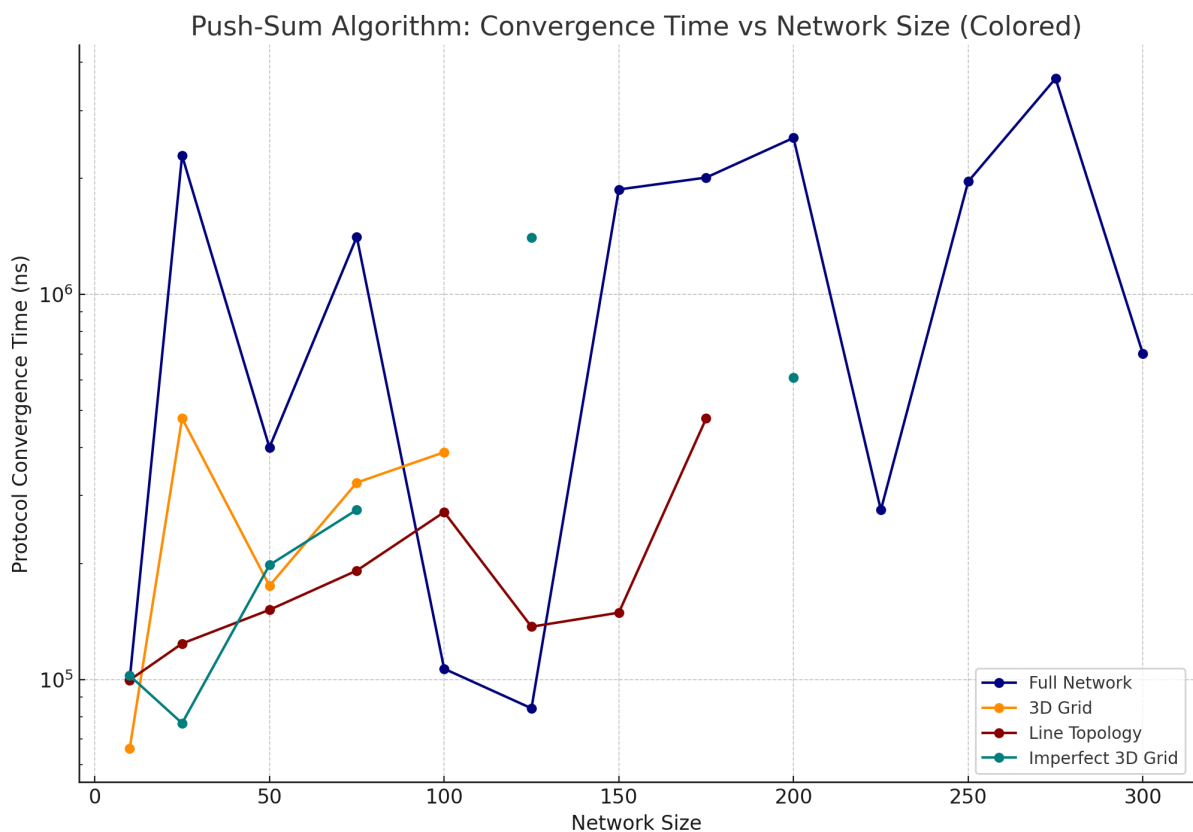
Line Topology:

Higher, steadily increasing convergence times due to sequential propagation, less efficient for larger networks.

Imperfect 3D Grid Topology:

Random additional connections reduce convergence time, offering shortcuts that improve efficiency without high complexity.

This is Push-Sum Algorithm Graph:



Findings from the Push-Sum Algorithm Graph:

Full Network Topology:

Convergence times vary significantly with network size due to complex, variable overhead. The growth is not linear, indicating challenges in maintaining full connectivity, which results in unpredictable convergence performance.

3D Grid Topology:

Shows a consistent trend with moderately higher but predictable convergence times. The structured layout results in stable communication patterns, leading to more uniform convergence behavior as network size grows.

Line Topology:

Displays higher convergence times due to sequential information propagation. Convergence

grows steadily but becomes substantially slower at larger network sizes, highlighting inefficiencies of the linear structure for large-scale networks.

Imperfect 3D Grid Topology:

Additional random connections provide beneficial shortcuts, improving efficiency compared to the 3D Grid. Convergence times are moderate, suggesting that imperfections enhance the spread without adding substantial complexity, resulting in faster convergence.